Physics-informed Neural Networks and properties of Quantum Dots Systems

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I. INTRODUCTION

Semiconductor quantum dot systems constitute one of the most versatile and controllable platforms for studying quantum mechanical many-body phenomena. Often referred to as artificial atoms, quantum dots allow for precise tuning of particle number, confinement geometry, interaction strength, and coupling to external fields through electrostatic gates and heterostructure design. This high degree of experimental control has positioned quantum dots at the intersection of condensed matter physics, quantum information science, and nanotechnology, enabling systematic investigations of correlation effects, coherence, and nonequilibrium dynamics at the nanoscale.

From a theoretical perspective, quantum dots provide a clean realization of interacting fermionic systems in reduced dimensions, where Coulomb interactions, exchange effects, and quantum confinement all play essential roles. Even for a small number of particles, the interplay between single-particle shell structure and many-body correlations leads to rich physical behavior, including Wigner molecule formation, spin ordering, and nontrivial excitation spectra. As the system size increases, the Hilbert space grows combinatorially, rendering exact solutions intractable and placing quantum dot systems squarely within the broader challenge of quantum many-body physics.

Traditional theoretical approaches such as exact diagonalization, configuration interaction, quantum Monte Carlo, density functional theory, and mean-field or perturbative methods, have each achieved notable success in specific regimes. However, these methods face intrinsic limitations. Exact diagonalization is restricted to very small particle numbers, Monte Carlo methods may suffer from the fermion sign problem, and approximate schemes often rely on uncontrolled assumptions that limit their predictive power in strongly correlated or nonequilibrium regimes. These challenges are exacerbated when one seeks to explore high-dimensional parameter spaces, optimize device geometries, or model realistic experimental conditions.

In recent years, machine learning and neural-network–based approaches have emerged as powerful complementary tools for tackling complex problems in quantum many-body physics. Neural networks offer flexible, high-dimensional function representations that can efficiently capture nontrivial correlations and emergent structures. Notable examples include neural-network quantum states, autoregressive wave-function models, and hybrid schemes combining variational principles with deep learning. These methods have demonstrated remarkable expressivity, often achieving accuracies comparable to or exceeding traditional variational ansätze.

Within this broader context, physics-informed neural networks (PINNs) provide a particularly promising framework for quantum dot studies. Unlike purely data-driven models, PINNs embed physical laws, symmetries, and constraints directly into the learning process, typically by incorporating differential equations, boundary conditions, conservation laws, and known operator structures into the loss function. For quantum systems, this allows the Schrödinger equation, many-body Hamiltonians, and symmetry requirements, such as antisymmetry under particle exchange, to guide the optimization of neural representations.

Applied to quantum dot systems, PINNs offer several compelling advantages. First, they enable the direct solution of many-body Schrödinger equations in continuous space without reliance on predefined basis sets, mitigating basis truncation errors. Second, they naturally accommodate complex geometries and external potentials, making them well suited for realistic device modeling. Third, by employing physical constraints, PINNs often require significantly less training data than conventional neural networks, improving interpretability and robustness. Finally, PINNs provide a unified framework capable of interpolating between few-body and many-body regimes, as well as between equilibrium and time-dependent dynamics.

The integration of physics-informed machine learning with quantum dot many-body theory thus represents a promising pathway toward overcoming longstanding computational barriers. Beyond providing efficient numerical solvers, these approaches open new avenues for discovering emergent structure, identifying effective degrees of freedom, and accelerating design and control of quantum devices. As quantum dot platforms continue to mature experimentally, the development of scalable, physically grounded theoretical methods will be essential for fully exploiting their potential in quantum simulation, quantum computation, and nanoscale electronics.

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by minimizing the expectation value of the Hamiltonian using Monte Carlo sampling. Prominent examples include restricted Boltzmann machines, feed-forward networks, and more recently deep convolutional or transformer-based architectures. VMC methods are firmly rooted in the variational principle and are particularly effective for ground-state problems. However, they rely on stochastic sampling in configuration space, which can become computationally

expensive in high dimensions or for fermionic systems with complex nodal structures. Moreover, the quality of the results depends sensitively on the chosen sampling strategy and the expressivity of the neural ansatz, and extensions to excited states or real-time dynamics remain challenging.

Autoregressive neural states (ANS) address some of these limitations by representing the wave function as a product of conditional probability amplitudes, enabling exact and unbiased sampling without Markov chains. This eliminates autocorrelation issues and significantly improves training stability. Autoregressive models have demonstrated impressive performance for lattice systems and discrete degrees of freedom, and they are particularly well suited for large-scale variational calculations. However, their application to continuous-space systems, such as quantum dots with realistic confinement potentials, is less straightforward and often requires nontrivial discretization or coordinate transformations. In addition, like VMC, autoregressive approaches remain fundamentally variational and typically focus on stationary states.

In contrast, physics-informed neural networks (PINNs) take a conceptually different approach. Rather than optimizing a variational energy functional via stochastic sampling, PINNs incorporate the governing physical equations directly into the loss function, enforcing the many-body Schrödinger equation, boundary conditions, and symmetry constraints pointwise in configuration space. This eliminates the need for Monte Carlo sampling altogether and allows the wave function to be learned as a continuous function of particle coordinates. For quantum dot systems, this is particularly advantageous, as it avoids basis truncation and naturally accommodates arbitrary confinement geometries and external fields.

Another key distinction lies in the treatment of fermionic antisymmetry and physical constraints. In VMC and autoregressive approaches, antisymmetry is typically enforced through carefully designed network architectures or determinant-based constructions. In PINNs, antisymmetry, normalization, and conservation laws can instead be imposed explicitly as constraints or penalty terms, providing a transparent and modular way to encode known physics. This often improves interpretability and reduces the effective hypothesis space explored during training.

From a computational perspective, the methods also differ in their scaling and scope. VMC and autoregressive neural states are generally more mature for large particle numbers and benefit from well-established stochastic optimization techniques. PINNs, while still an active area of research in quantum many-body physics, offer unique advantages for few- to intermediate-size systems, excited states, and time-dependent dynamics, where traditional variational approaches are less effective. Moreover, PINNs naturally lend themselves to hybrid strategies, where neural representations are guided simultaneously by data, variational principles, and physical laws.

In summary, VMC, autoregressive neural states, and PINNs should be viewed as complementary rather than competing approaches. VMC and autoregressive models excel as powerful variational solvers for ground-state problems, while PINNs provide a flexible, physics-driven framework that is particularly well suited for continuous-space quantum dot systems, complex geometries, and dynamical or constrained problems. The integration and cross-fertilization of these paradigms represent a promising direction for future advances in quantum many-body modeling.

II. THEORETICAL APPROACHES

III. PHYSICS-INFORMED NEURAL NETWORKS

Traditional machine learning focuses on fitting models to labeled data. In scientific applications, however, governing equations such as differential equations encode a wealth of prior information about the system. *Physics-Informed Neural Networks (PINNs)* integrate these constraints directly into the learning process, combining data-driven approximation with analytical structure. Figure **??** illustrates how the behaviour of neural networks changes when working with higher order derivatives, as is common in PINNs.

A. Formulation of PINNs

Let $u : \Omega \subset \mathbb{R}^d \to \mathbb{R}$ denote the unknown solution of a partial differential equation (PDE) of the form

$$\mathcal{N}[u](x) = 0, \qquad x \in \Omega, \tag{1}$$

subject to boundary conditions $u(x) = g(x)$ for $x \in \partial\Omega$. Here $\mathcal{N}$ is a (possibly nonlinear) differential operator.

In a PINN, $u_\theta(x)$ is represented by a neural network, and the loss function enforces both data fitting and PDE residual minimization:

$$\mathcal{L}(\theta) = \lambda_{\text{data}} \sum_{i=1}^{n_d} |u_\theta(x_i) - y_i|^2 + \lambda_{\text{PDE}} \sum_{j=1}^{n_r} |\mathcal{N}[u_\theta](\xi_j)|^2 + \lambda_{\text{BC}} \sum_{k=1}^{n_b} |u_\theta(\zeta_k) - g(\zeta_k)|^2. \tag{2}$$

Here $\{x_i, y_i\}$ are observed data, $\{\xi_j\}$ are collocation points in the domain for enforcing the PDE, and $\{\zeta_k\}$ are boundary points. The weights $\lambda_{\text{data}}, \lambda_{\text{PDE}}, \lambda_{\text{BC}}$ balance the different contributions.

B. Automatic differentiation

A key ingredient of PINNs is the use of *automatic differentiation* (AD) to evaluate ∇u_θ and Δu_θ. Unlike finite-difference approximations, AD computes derivatives exactly (up to floating-point error), making it feasible to incorporate higher-order differential operators directly into the loss. This enables end-to-end training of networks constrained by physical laws.

C. Theoretical difficulties

Although PINNs are conceptually straightforward, making them reliable and scalable raises several non-trivial challenges that go beyond classical universal approximation results.

a. Approximation in Sobolev spaces. Universal approximation theorems control function error in L^2/L^∞, but strong-form PINN losses depend on *derivatives* through $L[u_\theta]$. To make residuals small one must approximate in matching Sobolev norms $W^{\ell,p}$ (with ℓ the PDE order), which imposes smooth activations (e.g., tanh/SiLU for $\ell \geq 2$) and architecture choices that control higher derivatives. Low-regularity targets (discontinuities, cusps) break global $W^{\ell,p}$ control and favor weak/variational or entropy-based formulations.

b. Stability (residual-to-error). A small residual does not imply a small solution error unless the forward problem is (conditionally) stable. For well-posed elliptic/parabolic problems one can bound $\|u - u_\theta\|$ by $\|L[u_\theta] - f\|$, but inverse/ill-posed settings may require regularization, priors, or restricted hypothesis classes to prevent "residual overfitting" with large state error.

c. Sampling and generalization. Collocation-based training minimizes a sampled surrogate of a continuum loss. Generalization requires capacity control (e.g., bounded weights/Lipschitz loss) and sufficient sampling of interior vs. boundary/IC points; otherwise the empirical residual can be small while the population residual remains large.

d. Optimization and conditioning. PINN losses couple multiple physics terms (interior, boundary, data) with disparate scales, producing ill-conditioned gradient dynamics. In the linearized/NTK view, convergence speed is governed by the spectrum of a problem-dependent kernel; large condition numbers lead to slow or stalled training.

e. Spectral bias. Gradient-based training tends to fit low-frequency components first, delaying convergence for solutions with sharp layers or oscillations unless the sampling, features, or architecture emphasize high-frequency content (e.g., Fourier features, windowed bases, curriculum).

Mitigating these challenges requires careful design of architectures, activations, initialization, loss weighting, and sampling strategies. These topics are explained and discussed in detail in [**?**]

D. Implications for scientific machine learning

For physics-informed models, generalization takes on a slightly different meaning: the goal is not merely prediction on unseen data, but consistency with underlying physical laws. Inductive biases (symmetry, smoothness, conservation laws) are therefore not optional but essential. Implicit regularization from gradient-based optimization complements these biases by favoring solutions that are both smooth and physically plausible. Generalization in neural networks arises from an interplay of:

1. Algorithmic stability of gradient-based methods,
2. Implicit bias toward simple, smooth, low-complexity solutions, and
3. Architectural inductive biases encoding domain knowledge.

In scientific computing, these mechanisms explain why neural networks, despite their size, can approximate physically meaningful solutions. They also justify the careful design choices of activations, initialization, and constraints emphasized throughout this chapter.

Mitigation in practice. The challenges discussed for PINNs do not resolve themselves; they must be engineered away. In brief: use weak/variational formulations (or energy principles) when higher-order derivatives make strong forms unstable; hard–enforce essential boundary conditions and adapt weights for the rest; non-dimensionalize and

balance loss terms to improve conditioning; prefer smooth activations and architectures that control derivatives; exploit symmetry, equivariance, and locality to reduce effective complexity; precondition optimization with second-order or natural-gradient methods (e.g., L-BFGS, Stochastic Reconfiguration); and design sampling (importance/stratified, curricula in time/space) to close the generalization gap. Together, these measures turn inductive bias into concrete numerical stability and reliable, law-consistent generalization.

IV. OVERVIEW OF THE ANSATZ (AS IMPLEMENTED)

We use a Slater–Jastrow–neural correlator form, with optional backflow applied *only* to the coordinates entering the Slater determinant:

$$\Psi_{\theta,\beta}(R) = \mathrm{SD}\big(R'\big)\ \exp\Big(U(R) + W_\theta(R)\Big), \tag{3}$$

$$R' = R + \Delta_\beta(R). \tag{4}$$

Here SD enforces exact antisymmetry (see Sec. IV A); U is an explicit analytic pair term (Sec. IV C); W_θ is a compact permutation-invariant neural correlator (Sec. IV B); and Δ_β is an optional backflow displacement used only inside SD (Sec. IV D).

a. Coordinate conventions used by the correlator. In the implementation, the neural correlator operates on *trap-scaled* coordinates

$$\tilde{\mathbf{x}}_i = \sqrt{\omega}\,\mathbf{x}_i, \qquad \tilde{r}_{ij} = \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\|, \tag{5}$$

whereas the analytic pair term U (below) is evaluated on *unscaled* distances

$$r_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|. \tag{6}$$

Thus, W_θ depends on $\tilde{R}$ while U depends on R.

A. Slater reference

B. Neural correlator W_θ (implementation)

The correlator W_θ is a scalar function of the trap-scaled configuration $\tilde{R}$ built from (i) a particlewise branch (DeepSets) and (ii) a pairwise branch based on smooth "safe" distance features, followed by a small global readout. Pooling is always permutation-invariant.

1. *Particle branch (DeepSets).*

A network ϕ maps each particle $\tilde{\mathbf{x}}_i$ to a latent vector, and we average over particles:

$$\mathbf{h}_i^\phi = \phi(\tilde{\mathbf{x}}_i) \in \mathbb{R}^{d_L}, \qquad \overline{\mathbf{h}}^\phi = \frac{1}{N}\sum_{i=1}^{N} \mathbf{h}_i^\phi \in \mathbb{R}^{d_L}.$$

2. *Pair branch (safe features on unique pairs).*

We form unique pairs $i < j$ (not ordered pairs) and build features from the trap-scaled separation $\tilde{r}_{ij} = \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\|$. To keep first/second derivatives well-behaved at coalescence, we use a mollified radius

$$\tilde{r}_{ij}^{\mathrm{soft}} = \sqrt{\tilde{r}_{ij}^2 + \varepsilon^2}, \qquad \varepsilon = \varepsilon_{\mathrm{feat}}\, a_{\mathrm{ho}}, \qquad a_{\mathrm{ho}} = \omega^{-1/2}, \tag{7}$$

where $\varepsilon_{\mathrm{feat}}$ is a fixed dimensionless hyperparameter in code (e.g. $\varepsilon_{\mathrm{feat}} = 0.20$). Note that because $\varepsilon \propto a_{\mathrm{ho}}$, the effective mollification scale in the dimensionless $\tilde{r}$-geometry grows as $\omega^{-1/2}$ in the implementation.

From $\tilde{r}_{ij}$ and $\tilde{r}_{ij}^{\mathrm{soft}}$ we construct six smooth channels

$$s_1 = \log\Big(1 + (\tilde{r}_{ij}^{\mathrm{soft}}/\varepsilon)^2\Big), \qquad s_2 = \frac{\tilde{r}_{ij}^2}{\tilde{r}_{ij}^2 + \varepsilon^2}, \qquad s_3 = (\tilde{r}_{ij}^{\mathrm{soft}}/\varepsilon)^2 \exp\Big[-(\tilde{r}_{ij}^{\mathrm{soft}}/\varepsilon)^2\Big],$$
$$\mathrm{rbf}_g = \exp(-g\, s_1), \quad g \in \{0.25, 1, 4\}, \qquad \mathbf{u}_{ij} = [s_1, s_2, s_3, \mathrm{rbf}_{0.25}, \mathrm{rbf}_1, \mathrm{rbf}_4] \in \mathbb{R}^6. \tag{8}$$

Each pair feature vector is mapped by an MLP $\psi : \mathbb{R}^6 \to \mathbb{R}^{d_L}$ to a pair embedding $\mathbf{h}_{ij}^{\psi}$. We then apply a short-range gate *in the pair branch*

$$\chi(\tilde{r}_{ij}) = \frac{\tilde{r}_{ij}^2}{\tilde{r}_{ij}^2 + r_g^2}, \qquad r_g = r_{g,\mathrm{feat}}\, a_{\mathrm{ho}}, \tag{9}$$

with fixed dimensionless $r_{g,\mathrm{feat}}$ in code (e.g. $r_{g,\mathrm{feat}} = 0.30$), and average over unique pairs:

$$\overline{\mathbf{h}}^{\psi} = \frac{2}{N(N-1)} \sum_{i<j} \chi(\tilde{r}_{ij})\, \mathbf{h}_{ij}^{\psi} \in \mathbb{R}^{d_L}.$$

(As with ε, the effective gate radius in $\tilde{r}$-space scales $\propto a_{\mathrm{ho}}$ in the implementation.)

3. *Global "safe" extras and readout.*

In addition to pooled embeddings, we append two global scalars computed on trap-scaled coordinates:

$$g_1(\tilde{R}) = \frac{1}{Nd} \sum_{i=1}^{N} \|\tilde{\mathbf{x}}_i\|^2, \qquad g_2(\tilde{R}) = \frac{2}{N(N-1)} \sum_{i<j} s_1(\tilde{r}_{ij}), \tag{10}$$

and map everything to a scalar with a small MLP ρ:

$$W_\theta(\tilde{R}) = \rho\Big(\overline{\mathbf{h}}^{\phi} \parallel \overline{\mathbf{h}}^{\psi} \parallel [g_1(\tilde{R}), g_2(\tilde{R})]\Big). \tag{11}$$

C. Analytic pair term $U(R)$ (implementation)

Because the neural pair branch uses mollified/safe channels, we include an explicit analytic pair contribution in $\log \Psi$. In the implementation this term is evaluated on *unscaled* distances $r_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$:

$$U(R) = \sum_{i<j} u_{\sigma_i \sigma_j}(r_{ij}), \qquad u_{\sigma\sigma'}(r) = \gamma_{\sigma\sigma'}\, r\, e^{-r}. \tag{12}$$

Spin dependence is implemented via

$$\gamma_{\uparrow\downarrow} = \frac{1}{d-1}, \qquad \gamma_{\uparrow\uparrow} = \frac{1}{d+1}, \tag{13}$$

and $\gamma_{\sigma_i \sigma_j}$ is chosen based on whether the pair has equal spin ($\uparrow\uparrow$ or $\downarrow\downarrow$) or opposite spin. If no spin labels are provided, the code assigns a default half-up / half-down pattern.

D. Backflow (implementation)

Backflow is an optional nodal deformation applied only inside the Slater determinant,

$$R' = R + \Delta_\beta(R), \qquad \Psi_{\theta,\beta}(R) = \mathrm{SD}(R') \exp\big(U(R) + W_\theta(\tilde{R})\big).$$

In the codebase we use two backflow modules with the same interface $\Delta_\beta : \mathbb{R}^{N\times d} \to \mathbb{R}^{N\times d}$: a baseline message-passing model (`BackflowNet`) and a node/edge-feature variant (`CTNNBackflowNet`).

1. *Baseline* `BackflowNet`.

Given input coordinates $\mathbf{x}_i$ (in whatever units the wrapper supplies), we form pairwise geometry

$$\mathbf{r}_{ij} = \mathbf{x}_i - \mathbf{x}_j, \qquad r_{ij} = \|\mathbf{r}_{ij}\|, \qquad r_{ij}^2 = \|\mathbf{r}_{ij}\|^2,$$

and build per-edge messages with an MLP ϕ_β taking $(\mathbf{x}_i, \mathbf{x}_j, \mathbf{r}_{ij}, r_{ij}, r_{ij}^2)$:

$$\mathbf{m}_{ij} = \phi_\beta(\mathbf{x}_i, \mathbf{x}_j, \mathbf{r}_{ij}, r_{ij}, r_{ij}^2) \cdot w_{ij}^{\text{spin}} \cdot (1 - \delta_{ij}). \tag{14}$$

Here w_{ij}^{spin} is a binary mask (all pairs, or same-spin-only depending on configuration), and self-messages are removed. Messages are aggregated per node with a selectable reduction (`sum`/`mean`/`max`):

$$\mathbf{m}_i = \text{Agg}_j\, \mathbf{m}_{ij}.$$

A node/update MLP ψ_β produces displacements

$$\delta\mathbf{x}_i = \psi_\beta(\mathbf{x}_i \parallel \mathbf{m}_i) \in \mathbb{R}^d,$$

followed by an output bound and a positive scale:

$$\Delta_{\beta,i}(R) = s_\beta \tanh(\delta\mathbf{x}_i), \qquad s_\beta = \text{softplus}(s_\beta^{\text{raw}}) > 0. \tag{15}$$

To enforce center-of-mass neutrality we subtract the mean displacement:

$$\Delta_\beta \leftarrow \Delta_\beta - \frac{1}{N}\sum_{i=1}^{N} \Delta_{\beta,i}.$$

The final linear layer producing $\delta\mathbf{x}_i$ is zero-initialized so that $\Delta_\beta \simeq 0$ at initialization.

2. *Copresheaf-style* `CTNNBackflowNet`.

`CTNNBackflowNet` maintains explicit node features $\mathbf{h}_i \in \mathbb{R}^{H_v}$ and edge features $\mathbf{h}_{ij} \in \mathbb{R}^{H_e}$, with learned linear transports $\rho_{v\to e} : \mathbb{R}^{H_v} \to \mathbb{R}^{H_e}$ and $\rho_{e\to v} : \mathbb{R}^{H_e} \to \mathbb{R}^{H_v}$. In the implementation, this module first rescales its input internally as $\mathbf{x}_i \leftarrow \sqrt{\omega}\,\mathbf{x}_i$.

Node features are initialized by a linear embedding of $(\mathbf{x}_i)$ (optionally concatenated with a spin scalar), and edge features are initialized from the relative geometry $[\mathbf{r}_{ij}, r_{ij}, r_{ij}^2]$ via an MLP. One message-passing round is performed: (i) transport endpoint node features to edge space; (ii) update edges with an MLP; (iii) transport updated edges back to node space; (iv) aggregate incoming edge-to-node messages with `sum`/`mean`/`max`; (v) apply a residual node update MLP. A linear head maps updated node features to displacements, followed by the same output bound, COM projection, and positive scaling $s_\beta = \text{softplus}(s_\beta^{\text{raw}})$ as above, with the displacement head zero-initialized.

3. *Complexity.*

Both backflow variants construct dense pairwise interactions, so compute/memory scale as $O\big(BN^2(\cdot)\big)$ for batch size B (with the hidden widths determining the constant factors).

V. RESULTS AND DISCUSSIONS

A. Energy calculations

Table II reports ground-state energies obtained with residual-based pretraining followed by a stochastic reconfiguration (SR) tail, for both PINN+BF and PINN+CTNN trial states. Where available, we compare directly to diffusion Monte Carlo (DMC) reference values, and list the relative deviation (% err) from DMC.

TABLE I. Energy-partition diagnostics for the CTNN variational states used in the structural analysis. We list the interaction-to-kinetic ratio $\Gamma = \langle V_{\rm int}\rangle/\langle T\rangle$, the quasi-classical balance probe $\mathcal{C} = 2\langle V_{\rm trap}\rangle/\langle V_{\rm int}\rangle$, and the RMS radius $r_{\rm rms}$ extracted from $\langle V_{\rm trap}\rangle$ via Eq. (19), at the endpoints $\omega = 1$ and (where available) $\omega = 10^{-3}$.

N	$\omega = 1$			$\omega = 10^{-3}$		
	Γ	$\mathcal{C}$	$r_{\rm rms}$	Γ	$\mathcal{C}$	$r_{\rm rms}$
2	0.931	3.135	1.136	8.64	1.226	69.9
6	2.41	1.828	1.624	24.55	1.074	126
12	3.61	1.555	2.035	41.11	1.049	171
20	4.66	1.430	2.403	57.26	1.038	209

For $N = 2$, both ansätze reproduce the DMC benchmarks essentially at the level of the reported uncertainties across the full confinement range $\omega \in [0.01, 1]$. This indicates that, for the two-electron problem, the optimization pipeline is able to find a near-optimal nodal/shape representation for the ground state in all regimes considered.

For larger particle numbers, the comparison highlights a consistent performance difference between the two ansätze in the externally benchmarked subset. For $N = 6$ and $N = 12$, PINN+CTNN agrees more closely with DMC at moderate and strong confinement (e.g. $\omega = 0.1, 0.5, 1.0$), while PINN+BF exhibits small but systematic positive biases in several cases. For $N = 20$, DMC comparisons at $\omega \in \{0.1, 0.5, 1.0\}$ show that the CTNN ansatz remains accurate at scale, with the largest deviation appearing at $\omega = 0.1$ and substantially smaller deviations at $\omega = 0.5$ and $\omega = 1.0$.

For the weakest confinements ($\omega \in \{10^{-3}, 10^{-2}\}$) DMC references are not listed in Table II for several N, and these energies should therefore be interpreted as internally consistent variational results supporting the structural analysis below, rather than as externally benchmarked error statements. To quantify this internal consistency, we additionally analyze the energy partitioning

$$E \;=\; \langle T\rangle + \langle V_{\rm int}\rangle + \langle V_{\rm trap}\rangle, \qquad V_{\rm trap} = \tfrac{1}{2}\omega^2 \sum_{i=1}^{N} \langle r_i^2\rangle, \tag{16}$$

together with two diagnostics that directly probe the approach to the interaction-dominated (Wigner) regime:

$$\Gamma(\omega) \;=\; \frac{\langle V_{\rm int}\rangle}{\langle T\rangle}, \qquad \mathcal{C}(\omega) \;=\; \frac{2\langle V_{\rm trap}\rangle}{\langle V_{\rm int}\rangle}. \tag{17}$$

Across all reported (N, ω) the component estimators satisfy energy closure $E - (T + V_{\rm int} + V_{\rm trap}) \approx 0$ (typically 10^{-11}–10^{-7} in absolute units), and the harmonic-plus-Coulomb virial identity

$$2\langle T\rangle \;=\; 2\langle V_{\rm trap}\rangle - \langle V_{\rm int}\rangle \tag{18}$$

is satisfied to sub-percent relative accuracy (worst case $\lesssim 0.4\%$ in our dataset), providing a strong physics-based validation of the low-ω variational estimates.

The ratio trends are summarized in Table I and Fig. 1. As ω decreases, Γ increases sharply, e.g. from $\Gamma \simeq 0.93 \to 8.64$ for $N = 2$ and $\Gamma \simeq 4.66 \to 57.3$ for $N = 20$ when going from $\omega = 1$ to $\omega = 10^{-3}$, confirming a systematic crossover towards interaction-dominated states. Simultaneously, $\mathcal{C}(\omega)$ approaches unity at weak confinement, improving markedly with particle number: at $\omega = 10^{-3}$ we find $\mathcal{C} \simeq 1.226$ ($N = 2$), 1.074 ($N = 6$), 1.049 ($N = 12$), and 1.038 ($N = 20$), so that already at $N \geq 12$ the dot obeys $V_{\rm int} \approx 2V_{\rm trap}$ to within $\sim 5\%$ at $\omega = 10^{-3}$.

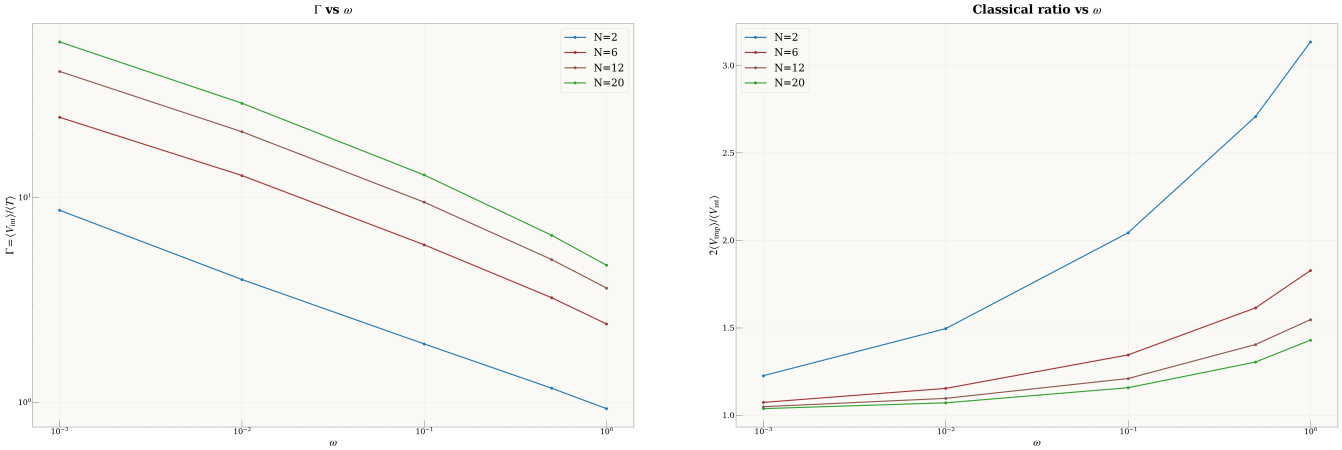
From $\langle V_{\rm trap}\rangle$ we also extract a root-mean-square radius per particle,

$$r_{\rm rms} \;=\; \sqrt{\frac{2\langle V_{\rm trap}\rangle}{N\omega^2}}, \tag{19}$$

which expands rapidly as confinement weakens (Fig. 1c). For $\omega = 1$ we obtain $r_{\rm rms} \approx 1.14, 1.62, 2.04, 2.40$ for $N = 2, 6, 12, 20$, respectively, whereas at $\omega = 10^{-3}$ the corresponding values are ≈ 70, 126, 171, 209, consistent with strong interaction-driven swelling in the Wigner regime.

B. Wigner regime: real-space ordering and mode-based validation

Having established energetic accuracy in the benchmarked regimes (Table II), we now turn to a structural question: *what configuration do the electrons adopt in real space as confinement weakens and Coulomb repulsion dominates?*



(a) $\Gamma(\omega) = \langle V_{\rm int}\rangle/\langle T\rangle$ (log–log).

(b) $\mathcal{C}(\omega) = 2\langle V_{\rm trap}\rangle/\langle V_{\rm int}\rangle$ (horizontal line: $\mathcal{C} = 1$).

FIG. 1. Energy-partition probes for the CTNN variational states. (a) Interaction dominance increases as confinement weakens. (b) The trap–Coulomb balance approaches the quasi-classical limit $\mathcal{C} \to 1$ at weak confinement, especially for larger N. (c) The cloud radius grows rapidly as $\omega \to 0$. (d) Component energies exhibit clean power-law scaling in ω on the studied grid.

TABLE II. Residual-based pretraining + SR tail. Reference DMC, PINN+BF and PINN+CTNN ground-state energies (Hartree) for selected (N, ω). Parentheses denote 1σ on the last digit(s). The % err columns are the relative deviations from DMC (%).

N	ω	Reference (DMC)	PINN+BF	% err	PINN+e	% err
2	0.001	—	0.0137948(8)	—	**0.013778(1)**	—
	0.01	0.073839(2)	0.073838(1)	−0.0014	0.0738382(5)	−0.0014
	0.10	0.44079(1)	0.44079(1)	+0.0000	0.440796(4)	−0.0000
	0.50	1.65977(1)	1.65976(2)	−0.0006	1.65976(1)	−0.0006
	1.00	3.00000	2.99999(1)	−0.0003	3.00000(2)	−0.0000
6	0.001	—	0.140913(1)	—	**0.1408172(7)**	—
	0.01	—	0.69125(2)	—	**0.69036(1)**	—
	0.10	3.55385(5)	3.5549(1)	+0.0295	**3.55388(5)**	+0.0008
	0.50	11.78484(6)	11.7895(4)	+0.0395	**11.7847(2)**	−0.0000
	1.00	20.15932(8)	20.1610(6)	+0.0083	**20.1585(3)**	−0.0041
12	0.001	—	0.515823(3)	—	**0.515365(4)**	—
	0.01	—	2.48620(5)	—	**2.47363(4)**	—
	0.10	12.26984(8)	12.2731(2)	+0.0160	**12.2718(1)**	—
	0.50	39.1596(1)	39.1786(8)	+0.0485	**39.1604(3)**	+0.0018
	1.00	65.7001(1)	65.717(1)	+0.0257	**65.69556(5)**	−0.0069
20	0.001	—	—	—	**1.293033(6)**	—
	0.01	—	—	—	**6.14645(5)**	—
	0.10	29.9779(1)	—	—	29.9888(2)	+0.0364
	0.50	93.8752(1)	—	—	**93.8789(5)**	+0.0040
	1.00	155.8822(1)	—	—	**155.8738(7)**	−0.0024

In a finite, rotationally symmetric trap the appropriate object is a *Wigner molecule*: electrons localize in *relative* coordinates (rings/shells with polygon-like angular correlations), while the entire pattern may rotate. Consequently, lab-frame angles can appear nearly uniform even when strong internal order is present. We therefore combine: (i) one-body density visualization, (ii) a label-invariant shell topology, (iii) rotation-registered bond-orientational order (internal order disentangled from lab-frame pinning), (iv) shell radii and separation gaps, (v) a pair-statistics validation isolating angular correlations beyond shelling, and (vi) a mode-alignment diagnostic testing whether the sampled fluctuations are consistent with structured small oscillations around a localized configuration.

a. One-body densities (visual crossover). Figures 2 and 3 compare one-body densities at strong confinement $\omega = 1.0$ and weak confinement $\omega = 0.001$ for $N = 2, 6, 12, 20$. At $\omega = 1.0$ the density remains compact and the

TABLE III. Key shell topologies (label-invariant). We report the occupancy pattern at the Wigner point $\omega = 10^{-3}$, the first post-Wigner step $\omega = 10^{-2}$, and strong confinement $\omega = 1$. The full ω grid is consistent with these transitions.

N	$\omega = 10^{-3}$	$\omega = 10^{-2}$	$\omega = 1$
2	(2)	(2)	(2)
6	(1,5)	(1,5)	(6)
12	(3,9)	(1,11)	(1,11)
20	(1,7,12)	(1,19)	(1,19)

TABLE IV. Shell radii and separation gaps at key confinement strengths (mean ± standard error over frames). Radii r_s are reported for populated shells; $\mathrm{gap}_{01} = \langle r_1 - r_0 \rangle$ and $\mathrm{gap}_{12} = \langle r_2 - r_1 \rangle$. Dashes indicate non-applicable shells/gaps.

N	ω	occ.	r_0	r_1	r_2	gap_{01}	gap_{12}
6	10^{-3}	(1,5)	35.56 ± 0.21	135.69 ± 0.06	–	100.13 ± 0.23	–
6	10^{-2}	(1,5)	13.348 ± 0.047	29.649 ± 0.026	–	16.301 ± 0.049	–
6	1	(6)	1.4996 ± 0.0023	–	–	–	–
12	10^{-3}	(3,9)	77.085 ± 0.074	189.420 ± 0.047	–	112.335 ± 0.092	–
12	10^{-2}	(1,11)	10.338 ± 0.031	37.662 ± 0.012	–	27.324 ± 0.028	–
12	1	(1,11)	0.5470 ± 0.0022	2.0020 ± 0.0015	–	1.4550 ± 0.0024	–
20	10^{-3}	(1,7,12)	38.04 ± 0.19	132.033 ± 0.038	247.226 ± 0.051	93.99 ± 0.18	115.193 ± 0.050
20	10^{-2}	(1,19)	10.315 ± 0.050	44.609 ± 0.012	–	34.294 ± 0.048	–
20	1	(1,19)	0.5547 ± 0.0022	2.3193 ± 0.0011	–	1.7646 ± 0.0021	–

dominant change with N is shell filling under the harmonic trap. At $\omega = 0.001$ the density broadens dramatically and develops pronounced radial structure for all N, indicating enhanced spatial separation and the onset of Coulomb-dominated localization.

b. Shell topology (discrete ring structure and where it collapses). To quantify radial ordering without relying on particle labels, we assign shells using a rank-based construction: at each frame, particles are sorted by radius $r_i(t) = \|\mathbf{r}_i(t)\|$ and grouped into shells by fixed rank cuts. The key qualitative information is the *occupancy pattern* and where it changes. Table III summarizes only the regimes that matter structurally: the deepest Wigner case $\omega = 10^{-3}$, the first post-Wigner step $\omega = 10^{-2}$ (where topology collapses for $N = 12$ and $N = 20$), and a representative strong-confinement point $\omega = 1$. At $\omega = 10^{-3}$ we recover the expected finite-N Wigner patterns: $(1, 5)$ for $N = 6$, $(3, 9)$ for $N = 12$, and a clear three-shell structure $(1, 7, 12)$ for $N = 20$. At $\omega = 10^{-2}$, $N = 12$ and $N = 20$ already collapse to $(1, 11)$ and $(1, 19)$, respectively, indicating a rapid loss of multi-shell localization as density increases. For $N = 6$ the two-ring pattern persists to $\omega \leq 0.1$ but is no longer robust at $\omega \geq 0.5$, where the rank-based boundary is not consistently separating and a single shell is detected.

c. Shell radii and separation gaps (quantitative localization). Topology tells *who sits on which ring*, but the Wigner regime is also characterized by large radii and well-separated shells. Table IV reports the mean radii of the populated shells and the inter-shell separation gaps at the same key points. In the Wigner regime ($\omega = 10^{-3}$) the shell radii become macroscopic on our scale: for $N = 12$ we find $r_0 \simeq 77.1$ and $r_1 \simeq 189.4$ with a large separation $\mathrm{gap}_{01} \simeq 112.3$, while $N = 20$ exhibits three shells at $r_0 \simeq 38.0$, $r_1 \simeq 132.0$, $r_2 \simeq 247.2$ with gaps $\mathrm{gap}_{01} \simeq 94.0$ and $\mathrm{gap}_{12} \simeq 115.2$. At $\omega = 10^{-2}$ the multi-shell structures for $N = 12$ and $N = 20$ have already collapsed into $(1, N-1)$, and the (single) separation gap_{01} drops accordingly, matching the topology change in Table III.

d. Rotation-registered angular order (internal order vs. lab-frame pinning). Shell occupancies capture *radial* localization but not whether particles on a ring exhibit polygon-like *relative* angular correlations. Because the Hamiltonian is rotationally symmetric, the correct diagnostic is a registered (co-rotating) frame: strong internal order may coexist with a diffusing global orientation. For shell s with n_s particles we compute

$$\psi_{n_s}^{\mathrm{reg}}(t) = \left| \frac{1}{n_s} \sum_{j \in s} e^{i n_s \theta_j^{\mathrm{reg}}(t)} \right|, \qquad \Phi_{s,\mathrm{reg}} = \langle \psi_{n_s}^{\mathrm{reg}} \rangle_t, \tag{20}$$

and summarize internal order by C-global $= \sum_s \Phi_{s,\mathrm{reg}}$ (which can exceed unity when multiple shells are ordered). We additionally report a lab-frame analogue $C_{\mathrm{global}} = \sum_s P_{s,\mathrm{lab}}$, which should be interpreted as a *pinning/rotation-timescale* diagnostic on the sampled time window.

TABLE V. Rotation-registered (internal) vs. lab-frame (pinning) angular order at key confinement strengths. C-global measures internal polygonal order in the registered frame, while C_{global} quantifies residual lab-frame pinning on the sampled time window. Uncertainties are standard errors over frames.

N	ω	occ.	C-global	C_{global}
2	10^{-3}	(2)	0.8746 ± 0.0012	0.0066 ± 0.0041
2	10^{-2}	(2)	0.7576 ± 0.0024	0.0066 ± 0.0034
2	1	(2)	0.6283 ± 0.0028	0.0034 ± 0.0020
6	10^{-3}	(1,5)	0.6520 ± 0.0020	0.0095 ± 0.0033
6	10^{-2}	(1,5)	0.4340 ± 0.0017	0.0041 ± 0.0022
6	1	(6)	0.3656 ± 0.0013	0.0042 ± 0.0018
12	10^{-3}	(3,9)	1.0684 ± 0.0043	0.0206 ± 0.0048
12	10^{-2}	(1,11)	0.2755 ± 0.0012	0.0046 ± 0.0015
12	1	(1,11)	0.2678 ± 0.0014	0.0032 ± 0.0013
20	10^{-3}	(1,7,12)	0.7909 ± 0.0027	0.0560 ± 0.0040
20	10^{-2}	(1,19)	0.2032 ± 0.0008	0.0004 ± 0.0007
20	1	(1,19)	0.2027 ± 0.0010	0.0030 ± 0.0010

TABLE VI. Registered Lindemann-type disorder at key confinement strengths (mean ± standard error over frames). For multi-shell Wigner states, both inner and outer shells are shown. Dashes indicate non-applicable shells.

N	ω	occ.	$L_{\text{in,reg}}$	$L_{\text{out,reg}}$
2	10^{-3}	(2)	–	0.1325 ± 0.0007
2	10^{-2}	(2)	–	0.2046 ± 0.0014
2	1	(2)	–	0.4004 ± 0.0022
6	10^{-3}	(1,5)	–	0.2049 ± 0.0008
6	10^{-2}	(1,5)	–	0.3443 ± 0.0011
6	1	(6)	–	0.5607 ± 0.0011
12	10^{-3}	(3,9)	0.2736 ± 0.0013	0.2547 ± 0.0010
12	10^{-2}	(1,11)	–	0.4500 ± 0.0010
12	1	(1,11)	–	0.6178 ± 0.0010
20	10^{-3}	(1,7,12)	0.3514 ± 0.0008	0.2696 ± 0.0007
20	10^{-2}	(1,19)	–	0.5716 ± 0.0011
20	1	(1,19)	–	0.6920 ± 0.0010

Table V shows that internal polygonal order is strongest in the Wigner regime and collapses rapidly when topology collapses. For $N = 12$, the Wigner case $\omega = 10^{-3}$ exhibits *two simultaneously ordered shells* and reaches C-global $= 1.0684 \pm 0.0043$, while already at $\omega = 10^{-2}$ (after the transition to $(1, 11)$) the internal order drops to 0.2755 ± 0.0012 and remains ≈ 0.268 at $\omega = 1$. For $N = 20$, $\omega = 10^{-3}$ again shows multi-shell internal order with C-global $= 0.7909 \pm 0.0027$, whereas the collapsed $(1, 19)$ states have C-global $\simeq 0.203$ at $\omega \geq 10^{-2}$. In all cases, C_{global} is much smaller than C-global, consistent with weak lab-frame pinning and substantial global rotation over the sampled trajectories.

e. Lindemann-type disorder (progressive melting with confinement). To complement angular order parameters we track a registered Lindemann-type measure $L_{s,\text{reg}}$ per shell, which quantifies the size of relative positional fluctuations compared to the shell scale (larger means more disordered). Table VI shows a consistent monotonic melting trend: weak confinement produces stiff rings, while strong confinement produces substantially larger relative fluctuations. For $N = 12$ and $N = 20$ the Wigner point $\omega = 10^{-3}$ exhibits *simultaneously stiff multiple shells* (e.g. $N = 12$: $L_{\text{in,reg}} = 0.2736 \pm 0.0013$, $L_{\text{out,reg}} = 0.2547 \pm 0.0010$; $N = 20$: $L_{7,\text{reg}} = 0.3514 \pm 0.0008$, $L_{12,\text{reg}} = 0.2696 \pm 0.0007$), while after collapse to $(1, N{-}1)$ the outer shell becomes increasingly disordered, reaching 0.6178 ± 0.0010 for $N = 12$ and 0.6920 ± 0.0010 for $N = 20$ at $\omega = 1$.

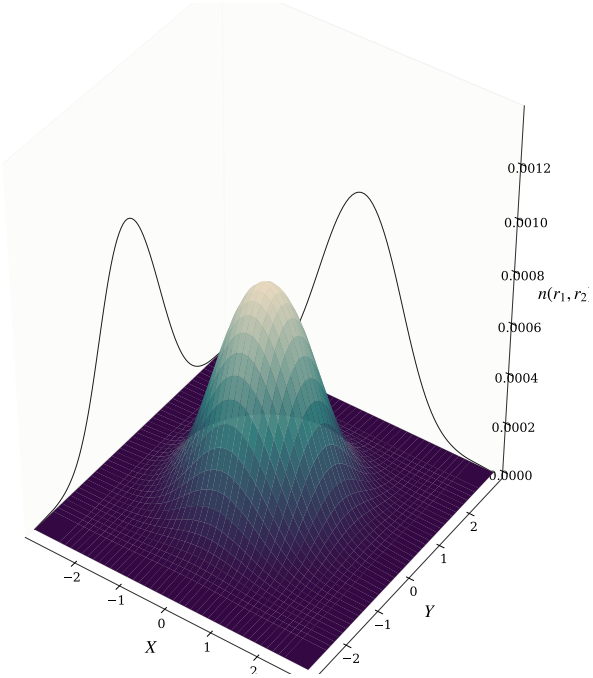
f. Mode-alignment diagnostic (are the fluctuations "molecule-like"?). A Wigner molecule is not merely a static ring pattern: if electrons are localized around a classical equilibrium, their residual motion should resemble structured small oscillations (collective distortions) rather than featureless fluid-like wandering. We test this by comparing

the empirical covariance C (center-of-mass removed) to the Hessian H of the corresponding classical equilibrium configuration (trap + Coulomb, scaled to the same ω). If fluctuations are organized by classical normal modes, C and H should be approximately compatible (near joint diagonalization) and a nontrivial fraction of variance should lie in the span of the softest modes. The diagnostics in Table **??** support this interpretation: for $N = 6$ the commutator norm is smallest at $\omega = 10^{-3}$ and increases with confinement, while for $N = 12$ and $N = 20$ the commutator norms remain small and overlap weights show a reproducible concentration of variance into a soft-mode subspace even though the fluctuation space is high-dimensional.

g. Pair-statistics validation (angular order beyond shelling). Shell topology captures the dominant radial organization, but it does not by itself guarantee polygon-like angular correlations. We therefore compare the empirical global pair-distance histogram to two shell-based generative models sharing the *same* shell assignment: (i) a polygon model (regular n_s-gons per shell with optimal phase alignment) and (ii) a uniform-angle model (same shell radii but no angular order). Table **??** shows that the polygon model systematically outperforms the uniform-angle baseline, and that the gain $\Delta\cos = \cos_{\text{poly}} - \cos_{\text{uni}}$ is strongest in the weakest trap and decreases with ω, isolating a genuine angular-correlation signal beyond radial shelling alone.

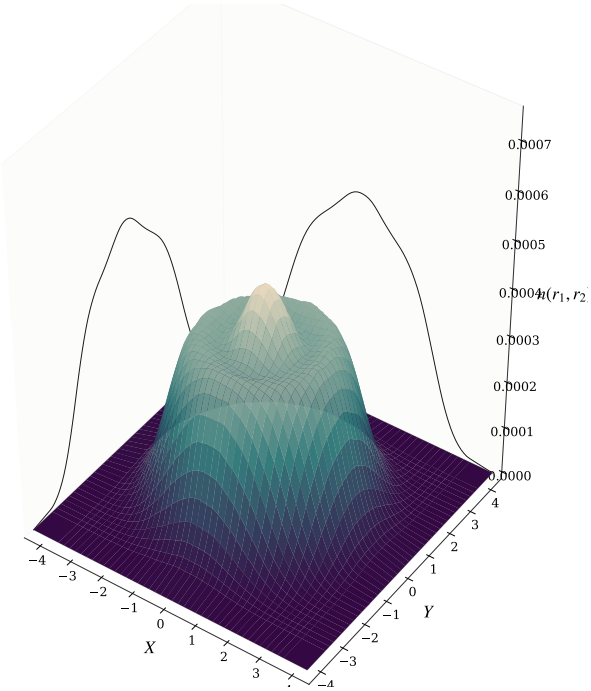
h. Discussion (what the combined diagnostics establish). Across $N = 2, 6, 12, 20$ we observe a consistent and internally validated Wigner-to-denser structural crossover as ω decreases. The one-body densities show dramatic spatial expansion and ring formation at $\omega = 10^{-3}$. The shell topology reveals *discrete* finite-N Wigner patterns at $\omega = 10^{-3}$ and sharp collapses at $\omega = 10^{-2}$ for $N = 12$ and $N = 20$. The shell radii and separation gaps provide a quantitative real-space counterpart: in the Wigner regime the shells are widely separated and stable, while their radii and gaps shrink rapidly as confinement strengthens. Rotation-registered bond-orientational order demonstrates strong polygon-like internal angular localization in the Wigner regime (including multi-shell ordering for $N = 12$ and $N = 20$), while the lab-frame measures remain comparatively small and should be interpreted as pinning/rotation-timescale diagnostics rather than internal order. The Lindemann measures rise strongly with confinement, giving an independent signature of progressive melting. Finally, the pair-distance validation confirms that angular correlations contribute explanatory power beyond shelling, with the largest gains at weak confinement, and the mode-alignment analysis adds mechanistic support: in the low-ω regime the fluctuations retain structured, collective content compatible with small-oscillation physics around localized configurations, as expected for Wigner molecules in a finite trap.

VI. CONCLUSION AND PERSPECTIVES

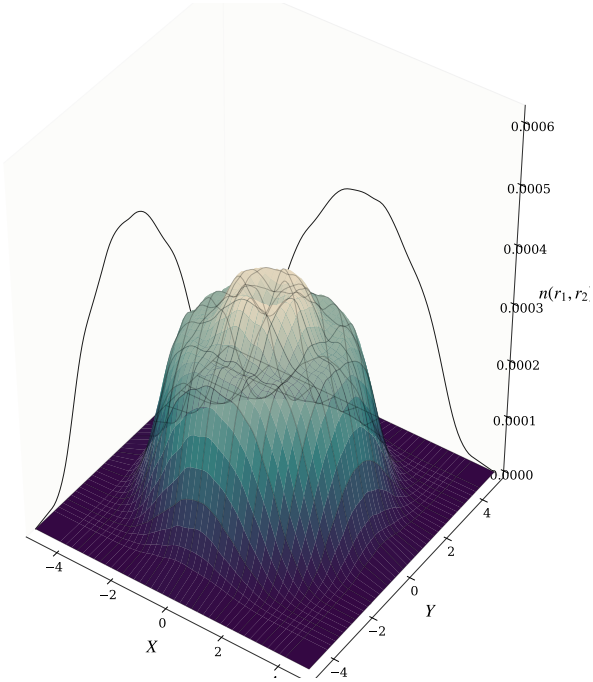


(a) N=2, $\omega = 1.0$

(b) N=6, $\omega = 1.0$

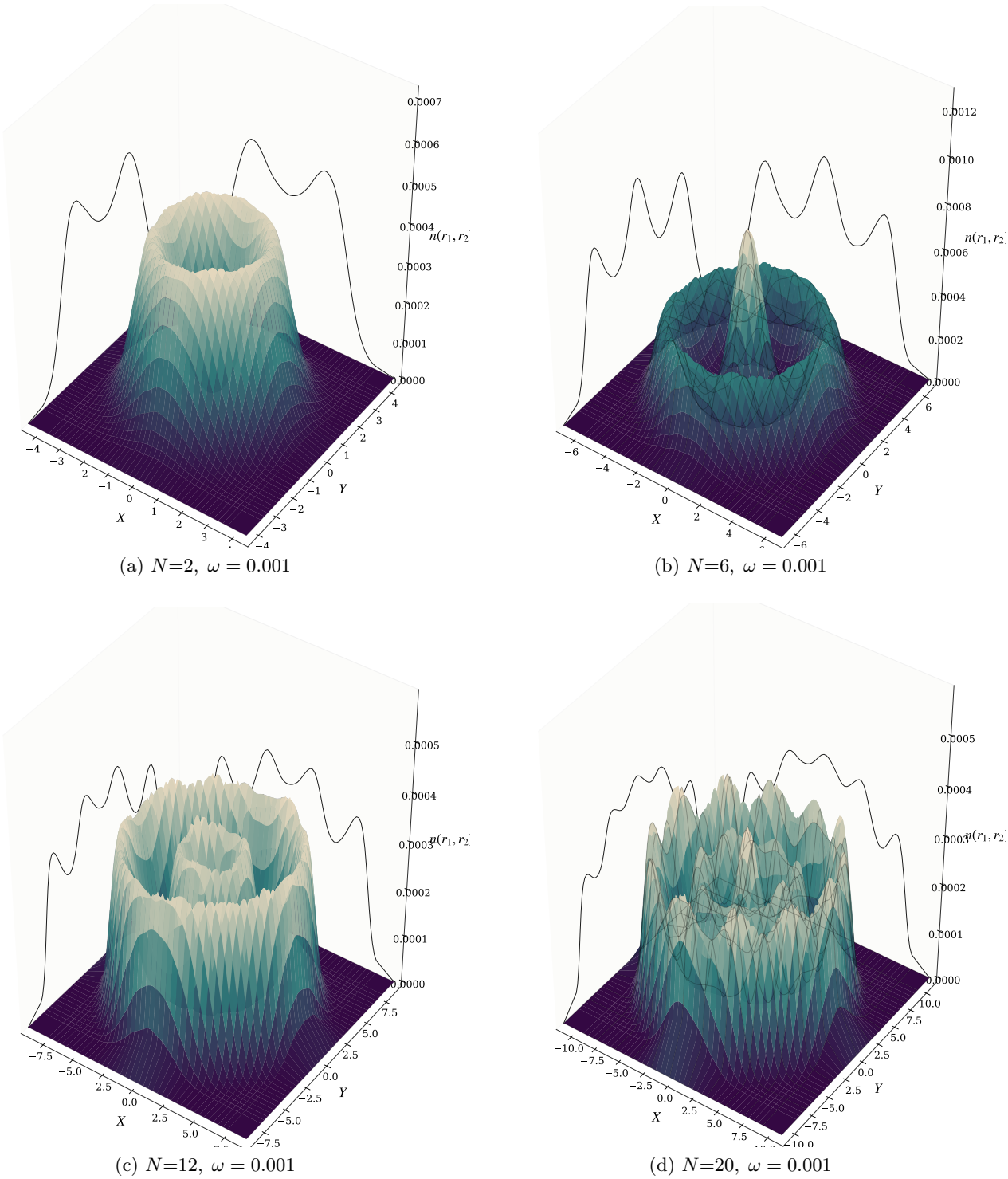


(c) N=12, $\omega = 1.0$



(d) N=20, $\omega = 1.0$

FIG. 2. One-body densities at strong confinement $\omega = 1.0$ for $N = 2, 6, 12, 20$. Increasing N primarily fills the trap while the overall extent remains compact due to the strong harmonic confinement.



(a) N=2, $\omega = 0.001$

(b) N=6, $\omega = 0.001$

(c) N=12, $\omega = 0.001$

(d) N=20, $\omega = 0.001$

FIG. 3. One-body densities at weak confinement $\omega = 0.001$ for $N = 2, 6, 12, 20$. The densities broaden and show strong radial structure, consistent with enhanced spatial separation and Wigner-molecule ordering in a Coulomb-dominated, low-density regime.